Amr Elsherif

Embedded Systems & ECE | IoT · Instrumentation & Control

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Education

Damanhour University — Sep 2022 – Jul 2027 expected
B.Eng. Electronics & Communications Engineering | GPA: 3.47/4.00 — *Damanhour, Egypt*

Experience

Network Engineering Trainee (CCNA) — Oct 2026 – Present
NTI, Digital Egypt Youth — *Damanhour, Egypt*
- Completing CCNA v7 training (preparing for 200-301); configured routers and switches in Cisco Packet Tracer labs covering VLANs, inter-VLAN, and static and OSPF routing.

Cloud Architect Intern (AWS) — Aug 2026
NTI × ITIDA (MCIT) — *Egypt*
- 120-hour internship; earned AWS Academy Graduate badges in Cloud Foundations and Cloud Architecting.

Instrumentation & Control Trainee — Jul – Aug 2026
Beheira Drinking Water & Wastewater Co. — *Damanhour, Egypt*
- Auto-tuned a PID temperature controller and installed it in a lab incubator, stable at a 100 °C setpoint.
- Tested AUMA valve-actuator control boards; calibration benches and PCB inspection under ESD safety rules.

Team Leader — Nov 2025 – Jul 2026
DEPI (MCIT), Harvest Training Center — *Alexandria, Egypt*
- Led the 20-trainee Cross-Platform Mobile group; coordinated teams and graduation-project follow-up.

Electrical Distribution Trainee — Jul – Aug 2025 (+2024)
Beheira Electricity Distribution Co. — *Damanhour, Egypt*
- Trained on medium- and low-voltage distribution networks, transformer substations, and safety compliance.
- Completed two summer rounds rated Excellent: 92% (2024) and 98% (2025).

Projects

Li-Fi Graduation Project | *Raspberry Pi 5, PPP, WireGuard, OpenCV* — Oct 2026 – Present
- Designing beam-steered Li-Fi with seamless Wi-Fi failover in a 4-member team (design phase).

DevSync | *Flutter, Firebase, Gemini API* | GitHub — May – Jul 2026
- Led the 5-member team that designed and built an AI app that matches developers with projects (Gemini API).

WeatherLink | *ESP32-S3, C++, PlatformIO, Blynk* | GitHub — Feb – May 2026
- Five-member team built a term project combining ESP32-S3 firmware, Blynk Cloud, and a Flutter app.

Ultrasonic Radar Mapping System | *PIC16F877A, Embedded C, HC-SR04, Servo*
- Built a 180-degree sweeping ultrasonic scanner using PIC hardware timers and PWM servo control.

Real-Time Audio Spectrum Analyzer | *Arduino Uno, C++, MAX9814, FHT*
- Built a real-time audio frequency visualizer using a Fast Hartley Transform (FHT) on the Arduino.

Skills

Embedded & IoT: ESP32, Raspberry Pi, PIC, Embedded Systems, IoT, Arduino, Proteus, PlatformIO
Instrumentation & Control: PID Controller Tuning, Calibration, Troubleshooting
Networking & Systems: TCP/IP, Subnetting, VLAN, OSPF, Cisco Packet Tracer, Linux, Git
Programming & Cloud: C, C++, Python, REST APIs, AWS (EC2, S3, VPC)

Certifications & Honors

AWS Academy Graduate – Cloud Architecting (credly.com/go/jEFPJNDz), Cloud Foundations (credly.com/go/pmZ02t1j); NTI Cloud Architect Internship (120 h); MCIT Certificate of Appreciation – Team Leader (Jul 2026); Official University Representative, EDEX 2025 (Dec 2025).